

```
-I/usr/lib/gcc/x86_64-redhat-linux/4.7.2/include \
-I/usr/include trun.c
```

```
trun.c:4:12: warning: cast truncates bits from constant value
(ffff becomes ff)
```

A truncation warning from Sparse for Linux-3.10.9 kernel is shown below:

```
arch/x86/kvm/svm.c:613:17: warning: cast truncates bits from
constant value (1000000000 becomes 0)
```

Any incorrect assignment between enums is checked with the `'-Wenum-mismatch'` option. To disable this check, use `'-Wno-enum-mismatch'`. Consider the following `enum.c` code:

```
enum e1 {a};
enum e2 {b};
```

```
int
main (void)
{
    enum e1 x;
    enum e2 y;

    x = y;

    return 0;
}
```

Testing with Sparse, you get the following output:

```
$ sparse -I/usr/include/linux \
-I/usr/lib/gcc/x86_64-redhat-linux/4.7.2/include \
-I/usr/include enum.c
```

```
enum.c:10:7: warning: mixing different enum types
enum.c:10:7:    int enum e2 versus
enum.c:10:7:    int enum e1
```

Similar Sparse warnings can also be seen for Linux 3.10.9:

```
drivers/leds/leds-lp3944.c:292:23: warning: mixing different
enum types
drivers/leds/leds-lp3944.c:292:23:    int enum led_
brightness versus
drivers/leds/leds-lp3944.c:292:23:    int enum lp3944_status
```

NULL is of pointer type, while the number 0 is of integer type. Any assignment of a pointer to 0 is flagged by the `'-Wnon-pointer-null'` option. This warning is enabled by default. An integer pointer 'p' is set to zero in the following

example:

```
int
main (void)
{
    int *p = 0;

    return 0;
}
```

Sparse notifies the assignment of 0 as a NULL pointer:

```
$ sparse -I/usr/include/linux \
-I/usr/lib/gcc/x86_64-redhat-linux/4.7.2/include \
-I/usr/include nullp.c
```

```
nullp.c:4:12: warning: Using plain integer as NULL pointer
```

Given below is another example of this warning in Linux 3.10.9:

```
arch/x86/kvm/vmx.c:8057:48: warning: Using plain integer as
NULL pointer
```

The corresponding source code on line number 8057 contains:

```
vmx->nested.apic_access_page = 0;
```

The GNU Compiler Collection (GCC) has an old, non-standard syntax for initialisation of fields in structures or unions:

```
static struct
{
    int x;
} local = { x: 0 };

int
main (void)
{
    return 0;
}
```

Sparse issues a warning when it encounters this syntax, and recommends the use of the C99 syntax:

```
$ sparse -I/usr/include/linux \
-I/usr/lib/gcc/x86_64-redhat-linux/4.7.2/include \
-I/usr/include old.c
```

```
old.c:4:13: warning: obsolete struct initializer, use C99
syntax
```